

Norway

SLEIPNER

Volve F

15/9-19

19 S

Design: 19 S

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	RT #2 @ 25.00m
Site:	Volve F	MD Reference:	RT #2 @ 25.00m
Well:	15/9-19	North Reference:	Grid
Wellbore:	19 S	Survey Calculation Method:	Minimum Curvature
Design:	19 S	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	15/9-19					
Well Position	+N/-S	6.89 m	Northing:	6,478,570.41 m	Latitude:	58° 26' 30.0331 N
	+E/-W	7.02 m	Easting:	435,057.04 m	Longitude:	1° 53' 15.3544 E
Position Uncertainty		0.00 m	Wellhead Depth:	84.00 m	Water Depth:	84.00 m

Wellbore	19 S				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MAGNETICREFERENCE	18.11.1999	-3.57	71.59	50,090

Design	19 S				
Audit Notes:					
Version:	10	Phase:	ACTUAL	Tie On Depth:	109.00
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	109.00	6.89	7.02	283.48	

Survey Program	Date	31.05.2010			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
122.95	1,492.92	19 SR RIGS 9 5/8" (19 SR)	RIGS, cont	Inertial Tool from BHI	
1,532.70	3,261.80	19 S MWD 12 1/4" (19 S)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
109.00	0.00	0.00	109.00	6.89	7.02	0.00	0.000	0.000	0.000	
122.95	0.07	171.75	122.96	6.84	6.99	0.02	0.151	0.151	0.000	
133.00	0.08	251.31	132.98	6.82	6.99	0.01	0.287	0.030	237.493	
142.96	0.08	250.95	142.96	6.81	7.01	-0.01	0.002	0.000	-1.084	
152.93	0.09	231.63	152.91	6.79	7.03	-0.03	0.091	0.030	-58.134	
162.98	0.04	245.05	162.96	6.75	7.07	-0.08	0.155	-0.149	40.060	
173.00	0.13	210.77	173.00	6.70	7.09	-0.11	0.298	0.269	-102.635	
182.94	0.06	189.59	182.92	6.62	7.10	-0.14	0.233	-0.211	-63.924	
192.92	0.18	247.61	192.92	6.55	7.08	-0.14	0.471	0.361	174.409	
202.92	0.18	250.26	202.91	6.52	7.05	-0.12	0.025	0.000	7.950	

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Well:	15/9-19	North Reference:	Grid
Wellbore:	19 S	Survey Calculation Method:	Minimum Curvature
Design:	19 S	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
212.94	0.15	237.42	212.93	6.53	7.02	-0.08	0.142	-0.090	-38.443
222.95	0.21	250.30	222.94	6.51	6.98	-0.05	0.216	0.180	38.601
233.00	0.21	261.96	232.99	6.51	6.94	-0.01	0.127	0.000	34.806
242.98	0.21	264.07	242.97	6.50	6.91	0.02	0.023	0.000	6.343
253.01	0.20	255.69	253.00	6.48	6.86	0.06	0.094	-0.030	-25.065
262.92	0.21	256.13	262.88	6.48	6.83	0.09	0.031	0.030	1.332
272.98	0.24	266.66	272.97	6.47	6.77	0.15	0.152	0.089	31.402
282.94	0.36	263.28	282.93	6.46	6.71	0.20	0.365	0.361	-10.181
293.00	0.45	263.01	292.98	6.47	6.64	0.27	0.268	0.268	-0.805
302.94	0.43	263.34	302.94	6.46	6.57	0.34	0.061	-0.060	0.996
313.00	0.47	258.62	312.98	6.46	6.50	0.41	0.163	0.119	-14.076
323.01	0.49	263.23	323.00	6.46	6.41	0.49	0.130	0.060	13.816
332.98	0.43	266.98	332.94	6.45	6.33	0.57	0.202	-0.181	11.284
342.92	0.48	262.99	342.91	6.44	6.24	0.65	0.179	0.151	-12.042
352.96	0.48	259.06	352.94	6.45	6.16	0.73	0.098	0.000	-11.743
363.02	0.50	258.20	363.00	6.46	6.08	0.81	0.064	0.060	-2.565
372.96	0.45	247.77	372.95	6.48	6.00	0.90	0.301	-0.151	-31.479
383.02	0.52	250.21	383.00	6.48	5.92	0.97	0.218	0.209	7.276
392.96	0.47	236.74	392.96	6.44	5.84	1.04	0.381	-0.151	-40.654
402.93	0.51	238.39	402.92	6.41	5.76	1.11	0.128	0.120	4.965
413.04	0.45	241.33	413.00	6.38	5.70	1.17	0.192	-0.178	8.724
422.93	0.40	232.42	422.92	6.34	5.64	1.21	0.251	-0.152	-27.027
432.93	0.42	231.74	432.92	6.32	5.57	1.28	0.062	0.060	-2.040
442.93	0.40	229.35	442.92	6.29	5.50	1.34	0.079	-0.060	-7.170
453.02	0.37	222.12	453.00	6.25	5.44	1.39	0.170	-0.089	-21.497
463.02	0.32	220.70	463.00	6.17	5.40	1.41	0.152	-0.150	-4.260
472.94	0.31	222.32	472.93	6.13	5.36	1.44	0.040	-0.030	4.899
482.95	0.31	221.66	482.93	6.08	5.31	1.47	0.011	0.000	-1.978
493.02	0.34	219.75	493.00	6.04	5.26	1.51	0.095	0.089	-5.690
503.02	0.39	210.74	503.01	5.99	5.22	1.54	0.228	0.150	-27.030
512.93	0.32	225.79	512.91	5.97	5.17	1.59	0.351	-0.212	45.560
523.00	0.33	212.91	522.97	5.93	5.12	1.62	0.219	0.030	-38.371
533.01	0.39	214.25	533.00	5.87	5.07	1.66	0.182	0.180	4.016
542.98	0.30	209.22	542.96	5.82	5.03	1.69	0.285	-0.271	-15.135
552.94	0.36	212.12	552.93	5.76	5.00	1.70	0.188	0.181	8.735
562.95	0.41	227.72	562.93	5.71	4.95	1.74	0.347	0.150	46.753
572.91	0.78	266.99	572.89	5.68	4.85	1.83	1.598	1.114	118.283
583.01	1.55	276.65	583.00	5.68	4.65	2.02	2.352	2.287	28.693
593.00	2.39	281.53	592.98	5.75	4.31	2.37	2.570	2.523	14.655
602.99	3.16	283.29	602.96	5.85	3.83	2.86	2.326	2.312	5.285
612.93	4.08	285.41	612.87	6.01	3.21	3.50	2.805	2.777	6.398
622.95	4.92	286.72	622.88	6.24	2.46	4.28	2.534	2.515	3.922
632.94	5.82	287.62	632.81	6.50	1.58	5.20	2.714	2.703	2.703

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Well:	15/9-19	North Reference:	Grid
Wellbore:	19 S	Survey Calculation Method:	Minimum Curvature
Design:	19 S	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
642.92	6.56	288.31	642.73	6.84	0.55	6.28	2.236	2.224	2.074
653.04	7.21	288.58	652.74	7.22	-0.60	7.49	1.929	1.927	0.800
663.02	7.77	288.65	662.67	7.65	-1.84	8.79	1.684	1.683	0.210
672.94	8.69	288.87	672.48	8.08	-3.18	10.21	2.784	2.782	0.665
683.01	9.52	288.85	682.43	8.61	-4.69	11.80	2.473	2.473	-0.060
692.96	10.22	289.34	692.25	9.19	-6.32	13.52	2.126	2.111	1.477
702.95	11.07	289.07	702.05	9.80	-8.09	15.38	2.557	2.553	-0.811
712.96	11.80	289.07	711.86	10.43	-9.96	17.35	2.188	2.188	0.000
722.92	12.67	288.61	721.59	11.13	-11.95	19.44	2.637	2.620	-1.386
732.95	13.47	287.95	731.37	11.83	-14.12	21.72	2.434	2.393	-1.974
743.00	14.53	287.49	741.11	12.58	-16.46	24.17	3.182	3.164	-1.373
753.02	15.57	287.46	750.78	13.35	-18.97	26.79	3.114	3.114	-0.090
762.99	16.72	287.40	760.37	14.19	-21.61	29.55	3.461	3.460	-0.181
772.97	17.75	287.84	769.89	15.11	-24.44	32.51	3.121	3.096	1.323
782.99	18.69	288.01	779.44	16.04	-27.44	35.65	2.819	2.814	0.509
793.01	19.66	287.78	788.87	17.01	-30.56	38.91	2.913	2.904	-0.689
802.96	20.57	288.08	798.23	18.07	-33.84	42.34	2.761	2.744	0.905
812.99	21.30	288.01	807.59	19.25	-37.20	45.89	2.185	2.183	-0.209
822.99	22.09	287.73	816.88	20.38	-40.71	49.56	2.390	2.370	-0.840
832.98	22.85	287.70	826.12	21.50	-44.38	53.39	2.283	2.282	-0.090
842.95	23.67	287.43	835.27	22.65	-48.18	57.35	2.488	2.467	-0.812
852.98	24.19	287.12	844.45	23.87	-52.07	61.42	1.600	1.555	-0.927
862.93	24.59	286.95	853.49	25.10	-55.94	65.48	1.224	1.206	-0.513
872.92	25.21	286.85	862.56	26.27	-59.94	69.64	1.866	1.862	-0.300
882.95	25.91	286.63	871.60	27.52	-64.12	74.00	2.113	2.094	-0.658
892.97	26.50	286.45	880.59	28.78	-68.36	78.41	1.782	1.766	-0.539
902.92	27.07	286.07	889.47	30.07	-72.65	82.88	1.794	1.719	-1.146
912.95	27.77	285.83	898.38	31.37	-77.08	87.50	2.120	2.094	-0.718
923.00	28.41	285.52	907.24	32.67	-81.66	92.25	1.959	1.910	-0.925
932.97	29.15	285.31	915.98	33.91	-86.31	97.06	2.247	2.227	-0.632
942.95	29.83	285.16	924.67	35.23	-91.05	101.98	2.056	2.044	-0.451
952.95	30.36	285.01	933.32	36.56	-95.93	107.03	1.606	1.590	-0.450
962.96	30.97	284.60	941.92	37.83	-100.91	112.17	1.933	1.828	-1.229
972.93	31.52	284.28	950.44	39.14	-105.93	117.36	1.729	1.655	-0.963
982.93	32.23	284.27	958.93	40.48	-111.04	122.65	2.130	2.130	-0.030
992.94	32.92	284.21	967.37	41.70	-116.29	128.04	2.070	2.068	-0.180
1,003.00	34.04	284.05	975.76	43.07	-121.68	133.60	3.350	3.340	-0.477
1,012.92	34.89	283.99	983.95	44.47	-127.14	139.23	2.573	2.571	-0.181
1,022.92	35.93	283.94	992.10	45.77	-132.75	144.99	3.121	3.120	-0.150
1,032.95	36.82	283.85	1,000.16	47.20	-138.51	150.92	2.667	2.662	-0.269
1,042.98	37.50	283.75	1,008.17	48.73	-144.39	157.00	2.042	2.034	-0.299
1,052.98	38.22	283.82	1,016.06	50.15	-150.36	163.13	2.164	2.160	0.210
1,062.93	39.31	283.78	1,023.83	51.66	-156.44	169.40	3.287	3.286	-0.121

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	RT #2 @ 25.00m
Site:	Volve F	MD Reference:	RT #2 @ 25.00m
Well:	15/9-19	North Reference:	Grid
Wellbore:	19 S	Survey Calculation Method:	Minimum Curvature
Design:	19 S	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
1,072.93	40.35	283.76	1,031.50	53.22	-162.67	175.82	3.120	3.120	-0.060	
1,082.95	41.42	283.72	1,039.07	54.77	-169.04	182.37	3.205	3.204	-0.120	
1,092.99	42.56	283.72	1,046.53	56.29	-175.62	189.13	3.406	3.406	0.000	
1,102.98	43.31	283.72	1,053.85	57.96	-182.21	195.93	2.252	2.252	0.000	
1,112.92	44.10	283.70	1,061.04	59.49	-188.86	202.75	2.385	2.384	-0.060	
1,122.98	45.14	283.72	1,068.20	61.07	-195.81	209.88	3.102	3.101	0.060	
1,132.92	46.24	283.67	1,075.14	62.73	-202.69	216.95	3.322	3.320	-0.151	
1,142.93	47.30	283.32	1,081.99	64.49	-209.81	224.29	3.267	3.177	-1.049	
1,152.94	48.34	283.11	1,088.72	66.22	-217.00	231.68	3.152	3.117	-0.629	
1,162.94	49.45	282.75	1,095.30	68.05	-224.30	239.21	3.428	3.330	-1.080	
1,172.96	50.48	282.55	1,101.74	69.59	-231.78	246.84	3.118	3.084	-0.599	
1,182.95	51.54	282.39	1,108.02	71.29	-239.35	254.60	3.205	3.183	-0.480	
1,192.97	52.49	282.27	1,114.19	73.00	-247.05	262.49	2.858	2.844	-0.359	
1,202.92	53.47	282.25	1,120.19	74.65	-254.81	270.42	2.955	2.955	-0.060	
1,212.95	54.43	282.17	1,126.10	76.35	-262.76	278.55	2.878	2.871	-0.239	
1,222.96	55.44	282.08	1,131.84	77.86	-270.75	286.67	3.035	3.027	-0.270	
1,232.96	56.26	282.01	1,137.47	79.73	-278.86	294.99	2.466	2.460	-0.210	
1,242.94	57.07	281.93	1,142.96	81.36	-286.99	303.27	2.443	2.435	-0.240	
1,252.95	57.44	281.90	1,148.36	83.09	-295.18	311.64	1.111	1.109	-0.090	
1,262.92	57.71	281.98	1,153.71	84.77	-303.45	320.07	0.837	0.812	0.241	
1,272.93	58.10	282.06	1,159.02	86.60	-311.75	328.57	1.186	1.169	0.240	
1,282.92	58.55	282.16	1,164.28	88.38	-320.04	337.06	1.375	1.351	0.300	
1,292.93	58.89	282.25	1,169.47	90.19	-328.39	345.60	1.045	1.019	0.270	
1,302.94	59.08	282.29	1,174.63	92.00	-336.71	354.11	0.579	0.569	0.120	
1,312.93	59.24	282.40	1,179.74	93.94	-345.09	362.71	0.558	0.480	0.330	
1,322.91	58.95	282.46	1,184.89	95.71	-353.44	371.24	0.885	-0.872	0.180	
1,332.92	58.46	282.45	1,190.09	97.53	-361.82	379.81	1.469	-1.469	-0.030	
1,342.91	58.05	282.34	1,195.34	99.29	-370.08	388.25	1.263	-1.231	-0.330	
1,352.96	57.74	282.42	1,200.68	101.12	-378.40	396.77	0.947	-0.925	0.239	
1,362.95	57.37	282.54	1,206.04	102.99	-386.61	405.19	1.152	-1.111	0.360	
1,372.94	56.77	282.59	1,211.48	104.85	-394.77	413.57	1.806	-1.802	0.150	
1,382.94	56.41	282.70	1,216.98	106.53	-402.94	421.90	1.115	-1.080	0.330	
1,392.93	55.92	282.83	1,222.54	108.38	-411.04	430.21	1.507	-1.471	0.390	
1,402.91	55.83	282.94	1,228.13	110.36	-419.14	438.55	0.385	-0.271	0.331	
1,412.91	56.29	283.03	1,233.74	112.07	-427.14	446.72	1.398	1.380	0.270	
1,422.94	57.06	283.03	1,239.26	114.11	-435.37	455.20	2.303	2.303	0.000	
1,432.95	57.71	283.09	1,244.64	116.08	-443.52	463.58	1.954	1.948	0.180	
1,442.92	57.80	283.07	1,249.96	117.96	-451.66	471.94	0.276	0.271	-0.060	
1,452.96	58.17	283.11	1,255.27	119.84	-459.96	480.45	1.110	1.106	0.120	
1,462.95	58.84	283.11	1,260.50	121.64	-468.30	488.98	2.012	2.012	0.000	
1,472.96	59.11	283.07	1,265.65	123.69	-476.67	497.60	0.816	0.809	-0.120	
1,482.92	58.79	283.21	1,270.76	125.70	-484.92	506.09	1.029	-0.964	0.422	
1,492.92	58.10	283.51	1,276.03	127.60	-493.22	514.60	2.208	-2.070	0.900	
1,532.70	60.30	284.14	1,296.40	135.76	-526.40	548.77	1.709	1.659	0.475	

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Site:	Volve F	MD Reference:	RT #2 @ 25.00m
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Wellbore:	19 S	Survey Calculation Method:	Minimum Curvature
Design:	19 S	Database:	Production EDM P246N

Survey									
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1,561.50	60.50	283.04	1,310.62	141.65	-550.74	573.81	1.018	0.208	-1.146
1,590.00	60.10	283.84	1,324.74	147.40	-574.82	598.57	0.844	-0.421	0.842
1,617.90	59.50	283.94	1,338.78	153.18	-598.22	622.68	0.652	-0.645	0.108
1,648.30	59.90	284.84	1,354.12	159.71	-623.65	648.92	0.862	0.395	0.888
1,675.90	59.70	282.34	1,368.00	165.31	-646.83	672.77	2.359	-0.217	-2.717
1,705.40	58.30	281.74	1,383.19	170.58	-671.56	698.05	1.517	-1.424	-0.610
1,732.30	58.50	284.24	1,397.29	175.73	-693.88	720.96	2.385	0.223	2.788
1,762.30	58.50	282.54	1,412.97	181.65	-718.77	746.54	1.449	0.000	-1.700
1,791.60	58.50	284.04	1,428.28	187.39	-743.08	771.52	1.310	0.000	1.536
1,821.20	58.60	282.64	1,443.72	193.22	-767.65	796.77	1.215	0.101	-1.419
1,850.30	58.30	282.14	1,458.95	198.54	-791.87	821.56	0.537	-0.309	-0.515
1,877.90	58.10	283.04	1,473.49	203.65	-814.76	845.01	0.859	-0.217	0.978
1,906.50	57.50	282.94	1,488.73	209.09	-838.35	869.22	0.636	-0.629	-0.105
1,935.10	57.70	283.74	1,504.06	214.66	-861.84	893.36	0.739	0.210	0.839
1,964.30	57.50	282.74	1,519.70	220.30	-885.84	918.02	0.891	-0.205	-1.027
1,993.60	57.60	282.84	1,535.43	225.77	-909.95	942.74	0.134	0.102	0.102
2,020.60	57.50	284.04	1,549.91	231.07	-932.11	965.52	1.131	-0.111	1.333
2,050.70	57.40	283.24	1,566.11	237.05	-956.77	990.89	0.679	-0.100	-0.797
2,079.40	57.60	283.44	1,581.53	242.63	-980.32	1,015.10	0.273	0.209	0.209
2,108.90	57.30	285.04	1,597.40	248.75	-1,004.42	1,039.96	1.405	-0.305	1.627
2,137.70	58.00	283.84	1,612.81	254.81	-1,027.98	1,064.29	1.283	0.729	-1.250
2,167.00	58.80	284.34	1,628.16	260.88	-1,052.19	1,089.24	0.928	0.819	0.512
2,196.10	58.70	283.94	1,643.26	266.96	-1,076.31	1,114.12	0.367	-0.103	-0.412
2,225.00	58.90	282.44	1,658.23	272.60	-1,100.38	1,138.84	1.348	0.208	-1.557
2,253.00	58.80	284.34	1,672.72	278.15	-1,123.69	1,162.80	1.745	-0.107	2.036
2,281.70	58.80	284.24	1,687.58	284.20	-1,147.48	1,187.34	0.089	0.000	-0.105
2,309.60	58.80	283.54	1,702.04	289.93	-1,170.65	1,211.21	0.644	0.000	-0.753
2,338.60	58.90	284.14	1,717.04	295.87	-1,194.75	1,236.03	0.541	0.103	0.621
2,367.00	58.90	283.54	1,731.71	301.68	-1,218.36	1,260.34	0.543	0.000	-0.634
2,395.70	58.90	283.64	1,746.53	307.45	-1,242.25	1,284.92	0.090	0.000	0.105
2,424.60	58.90	283.04	1,761.46	313.16	-1,266.32	1,309.66	0.533	0.000	-0.623
2,453.90	58.80	284.24	1,776.62	319.07	-1,290.69	1,334.74	1.056	-0.102	1.229
2,484.40	58.80	283.34	1,792.42	325.29	-1,316.03	1,360.83	0.757	0.000	-0.885
2,513.90	58.80	284.14	1,807.70	331.28	-1,340.54	1,386.06	0.696	0.000	0.814
2,541.00	58.90	284.34	1,821.72	336.99	-1,363.02	1,409.25	0.219	0.111	0.221
2,569.00	59.00	284.54	1,836.16	342.97	-1,386.25	1,433.24	0.213	0.107	0.214
2,597.00	58.90	284.14	1,850.60	348.91	-1,409.49	1,457.22	0.382	-0.107	-0.429
2,626.90	58.90	283.44	1,866.05	355.01	-1,434.36	1,482.82	0.601	0.000	-0.702
2,655.80	58.90	284.14	1,880.97	360.90	-1,458.39	1,507.57	0.622	0.000	0.727
2,684.60	58.00	282.34	1,896.04	366.52	-1,482.28	1,532.11	1.853	-0.937	-1.875
2,713.60	57.40	282.84	1,911.54	371.86	-1,506.20	1,556.62	0.759	-0.621	0.517
2,743.10	57.60	282.44	1,927.39	377.31	-1,530.48	1,581.50	0.399	0.203	-0.407
2,771.40	57.60	282.44	1,942.56	382.45	-1,553.81	1,605.39	0.000	0.000	0.000

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	RT #2 @ 25.00m
Site:	Volve F	MD Reference:	RT #2 @ 25.00m
Well:	15/9-19	North Reference:	Grid
Wellbore:	19 S	Survey Calculation Method:	Minimum Curvature
Design:	19 S	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,800.60	57.70	280.84	1,958.18	387.43	-1,577.97	1,630.04	1.392	0.103	-1.644	
2,829.50	57.80	282.94	1,973.60	392.46	-1,601.89	1,654.47	1.847	0.104	2.180	
2,857.90	57.80	282.94	1,988.74	397.84	-1,625.31	1,678.50	0.000	0.000	0.000	
2,886.70	57.90	282.84	2,004.06	403.28	-1,649.08	1,702.88	0.136	0.104	-0.104	
2,914.80	58.00	282.34	2,018.97	408.47	-1,672.32	1,726.70	0.465	0.107	-0.534	
2,943.40	57.90	280.94	2,034.15	413.36	-1,696.07	1,750.92	1.249	-0.105	-1.469	
2,971.90	57.70	281.44	2,049.34	418.04	-1,719.72	1,775.02	0.493	-0.211	0.526	
2,999.20	57.80	282.54	2,063.91	422.83	-1,742.31	1,798.10	1.028	0.110	1.209	
3,029.60	58.20	281.84	2,080.02	428.27	-1,767.51	1,823.87	0.706	0.395	-0.691	
3,058.80	57.80	283.44	2,095.49	433.69	-1,791.67	1,848.63	1.453	-0.411	1.644	
3,087.60	58.00	282.44	2,110.80	439.15	-1,815.45	1,873.03	0.907	0.208	-1.042	
3,114.80	58.20	282.54	2,125.17	444.14	-1,837.99	1,896.12	0.240	0.221	0.110	
3,144.00	58.30	284.84	2,140.54	450.02	-1,862.12	1,920.95	2.012	0.103	2.363	
3,174.20	58.00	285.54	2,156.47	456.74	-1,886.87	1,946.59	0.662	-0.298	0.695	
3,204.50	57.50	284.34	2,172.64	463.34	-1,911.63	1,972.20	1.120	-0.495	-1.188	
3,233.00	57.20	284.34	2,188.02	469.28	-1,934.88	1,996.20	0.316	-0.316	0.000	
3,261.80	57.50	285.34	2,203.56	475.49	-1,958.32	2,020.44	0.931	0.312	1.042	
3,580.00	57.50	285.34	2,374.52	546.47	-2,217.13	2,288.67	0.000	0.000	0.000	

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
175.00	175.00	30"	30.000	36.000	
535.00	534.99	20"	20.000	26.000	
1,485.00	1,271.84	13 3/8"	13.375	17.500	
3,569.30	2,368.77	9 5/8"	9.625	12.250	

Checked By: _____ Approved By: _____ Date: _____